MIL-T-81078(OS)
Amendment 6
16 April 1969
Superseding
Amendment 5

11 July 1968

MILITARY SPECIFICATION

TEST SET, AFTERBODY, TORPEDO MARK 442 MOD 0

(For Torpedo Mk 46 Mod 0)

This amendment forms a part of Military Specification MIL-T-81078(OS) of 26 May 1964 and has been approved by the Naval Ordnance Systems Command, Department of the Navy

By this amendment, MIL-T-81078(WP) of 26 May 1964 is changed to read MIL-T-81078(OS).

Wherever it appears in specification delete "Bureau of Naval Weapons" and substitute "Naval Ordnance Systems Command".

Page 5, paragraph 3.3.5. Delete the last two sentences and substitute: "The AC VOLTS signal lamp shall light when 23.5 plus or minus 2 volts root-mean-square (vrms) alternating current at 2000 plus or minus 100 cps is applied to pins J1-22 and J1-28. The lamp shall remain lit up to 40 plus or minus 2 vrms. The WATER PUMP lamp shall light when an external switch is closed between pins J5-P and J5-N."

Page 6, Table II: Under "Meter Reading" delete "36.0 ± 1.8 vdc" and substitute "38.4 ± 2.7 vdc".

Page 7. Add the following new paragraph:
"3.3.6.6 Relay K6 operation. When the TEST TYPE switch is set for STATIC and the STATIC TESTS test mode switch is set for MANUAL, the voltage measured between J3-10 and J3-1 shall be 6.3 vac; depressing the ADVANCE switch shall cause the voltage between J3-10 and J3-1 to go to zero."

Page 9, Table V: Under "Signal (vdc)" delete "34.0 ± 2.8" and substitute "36.4 ± 2.7" in three places.

Page 10, Table VI: Under "Signal Voltage" delete "34 ± 2.8" and substitute "36.4 ± 2.7" in eight places; delete "24 ± 2.8" and substitute "36.4 ± 2.7"; delete entry under "Limits" opposite Test Number 10 and substitute "430 ± 10 ms
368 ± 10 ms

FSC 1355

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Page 11. Add the following new paragraph:
"3.3.7.4 When the response simulator (4.7.1(h)) is connected to J-5 and is simulating a roll to port; voltage waveform measured between J6-19 and J6-1 shall be a sawtooth 10 to 20 volts in amplitude with a time period of approximately 250 milliseconds. When a roll to starboard is simulated, the voltage between J6-19 and J6-1 shall drop to zero, rise, and again drop to zero, and rise when a roll to port is again simulated: Finally, when the ADVANCE button is depressed, voltage should drop to zero."

Page 12, Table VIII: Under "Connector Pin" column delete "J5-J", "J5-K", "J5-L", and "J5-M".

Page 12, paragraph 3.3.9, line 4: Delete "pin J6-B" and substitute "pin J6-3".

Page 13, paragraph 3.4.4, line 2: Delete "vibration requirements" and substitute "Type I vibration requirements".

Page 13, paragraph 3.4.4.1, line 3: Delete "an amplitude" and substitute "a table amplitude" in two places.

Page 19, paragraphs 4.6, 4.6.1, and 4.6.2. Delete and substitute:
"4.6 Periodic production tests. The periodic production sample, selected in accordance with 4.3.2, shall be subjected to the periodic production test (4.6.2). The periodic production sample shall then be subjected to and satisfactorily pass the acceptance tests and examinations (4.7). Certified inspection data covering the results of the periodic production tests shall be forwarded to the Naval Ordnance Systems Command. Certified inspection data shall consist of the following:

(a) Items 4.5(a) through 4.5(c)

(b) Test results of periodic production tests (4.6.2)

4.6.1 Periodic production test equipment. The following items of test equipment, or equivalent items, will be required to perform the periodic production tests:

(a) Temperature chamber (see 4.5.1(j))

(b) Vibration test fixture (see 4.5.1(h))

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4.6.2 Periodic production tests. Subject the periodic production sample to the following tests:

(a) Temperature (4.5.2.1)

(b) Vibration (4.5.2.4)"

Page 20, paragraph 4.7.3.2(b): Delete "-36 ± 1.8" and substitute "-38.4 ± 2.7".

Page 20, paragraph 4.7.3.3(a). Delete and substitute:
"4.7.3.3(a) Connect audio oscillator (4.7.1(f)) between pins J1-22 and J1-28. Use an isolation power plug adapter. On oscillator, set frequency to 2000 ± 100 cps and output level to 18 vac rms. AC VOLTS lamp on test set shall not light."

Page 21, paragraph 4.7.3.3(b): Delete "32 vac" and substitute "26 vac".

Page 21, paragraph 4.7.3.3(c): Delete "Disconnect oscillator" and substitute "Increase output level to 40 ±2 vac rms; the AC VOLTS lamp shall remain lit."

Page 21. Add the following new paragraph:
"4.7.3.3(d) Disconnect oscillator."

Page 21, paragraph 4.7.3.6, line 4: Delete "10.7 ± 1.2 vdc" and substitute "12.5 ± 0.5 vdc".

Page 23, Table X: Under "Resistance" column for Test 26,
delete "1.2 and substitute "2.0
0.8" 0.5"

Page 24, paragraph 4.7.5.1(b): Delete "34.0 ± 2.8" and substitute "36.4 ± 2.7".

Page 24, paragraph 4.7.5.1.1, lines 1 and 2: Delete "between J1-42 and J1 48" and substitute "to J1-42(-) and J1-48(+)".

Page 25, paragraph 4.7.5.3(a): Delete first sentence and substitute "Connect 350 ohm ±5 percent resistors across pins J1-6 and J1-24, J1-12 and J1-24, and J1-18 and J1-24."

Page 27, paragraph 4.7.5.6. Delete and substitute:
"4.7.5.6 Time-to-failure-durations. On counter (4.7.1(e)), set COM/SEP switch to SEP, START POLARITY to (-), and STOP to (-).

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CAUTION

Damage to -36 vdc power supply may result if TRIGGER INPUT, of the counter, is left in the COM position while test leads are connected. Connect counter STOP jack to TP-2 on test set timing module and START jack to TP-3 on test set. Advance TEST NUMBER to 01."

Page 27. Add the following new paragraphs:

"4.7.5.8 Relay K6 operation. Remove clip lead from R43 and ground. Connect sensing circuit module to an extender cable and connect extender cable to Test Set J3. Connect oscilloscope between J3-10 and J3-1 ground. Set TEST TYPE to STATIC and STATIC TESTS test mode to MANUAL. Trace should indicate 16.0 to 19.5 volts peak-to-peak. Press ADVANCE button and note that voltage drops to zero momentarily.

4.7.5.8.1 Set TEST TYPE to FUNCTIONAL and FUNCTIONAL TESTS test mode to MANUAL. Remove extender cable from Sensing Circuity Module. Connect extender cable between test set and Timing Assembly Module A4. Connect oscilloscope between J6-19 and J6-1 (Ground). Connect response simulator (4.7.1(h)) to J5, set selector switch to ROLL, and press UP/PT/OPEN. Advance TEST NUMBER to 01 and note that oscilloscope trace displays a sawtooth wave of between approximately 10 to 20 volts at approximately 250 milliseconds period. Press DOWN/STBD/CLOSE button and note that trace drops to near zero then rises again. Then press UP/PT/OPEN button and notice that oscilloscope trace again drops to near zero and then rises again."

4.7.5.8.2 Advance test number to 02 while holding UP/PT/OPEN switch depressed, and note that oscilloscope trace again drops to near zero volts and rises back to sawtooth waveshape as above."

Page 27, paragraph 5.1.1: In line 4, delete "Process D-1 or D-4" and substitute "Procedure D-1"; in line 5, delete "Preservation and packaging shall be in accordance with Method IId."

Page 28, paragraph 5.2.1, line 3: Delete "Type I, class 2" and substitute "Type CF, Class, Weather resistant".

Page 31. Replace Figure 1 with new Figure 1, attached.

APPENDIX page ii, paragraph 10.3.2(c): Delete "36 ±1.8" and substitute "38.4 ± 2.7".

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APPENDIX page ii, paragraph 10.3.2(d): Under "Limits" column delete "10.3.2(b) ±1vdc" and substitute "10.3.2(c) ±1vdc"

APPENDIX page ii. Under paragraph 10.3.3 add the following sub-entry:
"(c) AC VOLTS lamp remains lit 4.7.3.3.(c) _____"

APPENDIX page iii, paragraph 10.3.6(a): Under "Limits" column delete "10.7 ± 1.2 vdc" and substitute "12.5 ± 0.5 vdc".

APPENDIX page iii, paragraph 10.3.6(b): Under "Inspection" column delete "FUNCTION MONITOR METER voltage" and substitute "FUNCTION MONITOR METER current".

APPENDIX pages iii and iv, paragraphs 10.3.8 (a), (b), (c), and (d). Delete entries under "Reference" and "Limits" columns and substitute:

"4.7.5.1 (a)	0.0 + 0.4 - 0.0 vdc
4.7.5.1 (b)	36.4 ± 2.7 vdc
4.7.4.1.1 (a)	6 ± 0.6 vdc
4.7.5.1.1 (b)	-6 ± 0.6 vdc"

APPENDIX page iv, paragraph 10.3.10(a). Under "Reference" column delete "4.7.5.3(a)" and substitute "4.7.5.3(b)".

APPENDIX page v. Under the "Limits" column of paragraph 10.3.10(j) delete "350 + 35 - 0 and substitute "430 ± 10
410 + 41 - 0 368 ± 10

APPENDIX page vi, paragraphs 10.3.14(a) and (b). Under "Reference" column add "4.7.5.7 in two places.

APPENDIX page vi. Add the following new paragraph:

"Inspection	Reference	Limits	Reading
10.3.15 Relay K6 Operating			
(a) Voltage (16.0 to 19.5 volts pk-pk)	4.7.5.8		_____
(b) Voltage drop	4.7.5.8		_____
(c) Sawtooth Shape (10 to 20v)	4.7.5.8.1		_____
(d) Voltage drop (DOWN/STBD/CLOSE)	4.7.5.8.1		_____

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10.3.15 (Cont'd)	Reference	Limits	Reading
(e) Voltage drop (UP/PT/OPEN)	4.7.5.8.1		_____
(f) Voltage drop (ADVANCE)	4.7.5.8.2		_____

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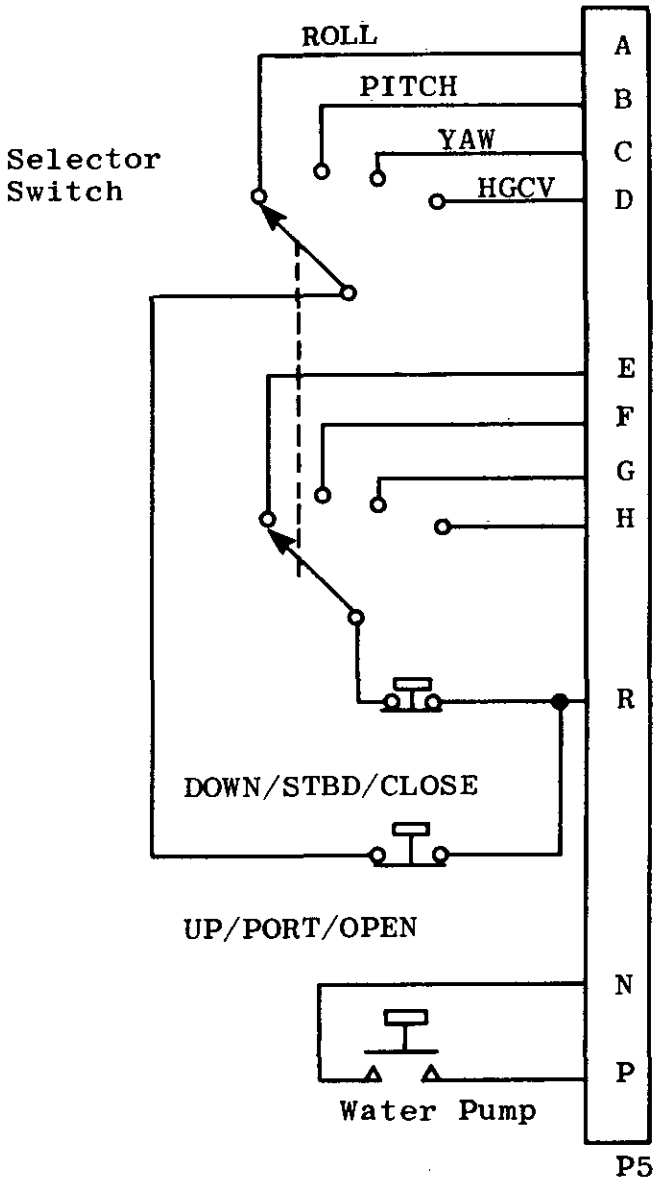


Figure 1. Response Simulator

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